Metropolitan Police Service
Live Facial Recognition Trials



National Physical Laboratory

Metropolitan Police Service

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1 Glossary of Terms

The terms defined within this section and as used throughout this report apply to this joint report only. The terminology and definitions may not apply to those adopted elsewhere by the MPS.

Alert: A notification of a possible match between a facial image from an individual present, and a facial image on the watchlist, where the comparison score exceeds the specified threshold.

Adjudication: The process of human assessment of an alert to decide whether to engage further with the individual matched to a watchlist image.

Bluelist: A partition of the watchlist containing facial images of police personnel (officers and/or staff) for use in the setup and evaluation of LFR comprising a set of subjects with controlled presence in the Zone of Recognition.

Live Facial Recognition (LFR): Real-time, automated facial recognition using video surveillance cameras.

False Negative Identification Rate (FNIR): The proportion of recognition opportunities of subjects who are on the watchlist that do not generate a correct alert. FNIR effectively indicates the number of [subjects] `missed' by the LFR system. The complement of FNIR is the **True Positive Identification Rate**

False positive alert: An alert that is not confirmed as a correct match for a subject's true identity. In this report false positive alerts include the following cases: alerts refuted by corroborative police checks, alerts dismissed in adjudication, and cases where requested engagement with the subject failed.

False positive identification rate (FPIR): The frequency of false positive alerts among recognition opportunities for individuals not included in the watchlist. Note that, in the operational context, it can be assumed that only a very small proportion of recognition opportunities will be for individuals on the watchlist.

IC code: Visual assessment ethnicity code, used by the MPS to record the perceived ethnicity of people [1]:

IC1: White – North European
IC2: White – South European
IC3: Black
IC4: Asian – Indian subcontinent
IC5: Chinese, Korean, Japanese, or other Southeast Asian
IC6: Arab or North African
IC9: Unknown

Assessment of IC codes may be somewhat subjective; different observers may sometimes assign a different IC code to the same individual. IC codes do not necessarily correspond to self-defined ethnicity and/or declared ethnicity.

Recognition Opportunity: The period when a person's face is visible to an LFR camera as they move through the Zone of Recognition.

True positive alert: An alert confirmed to be a correct match with a subject's true identity through satisfactory corroboration.

True positive identification rate (TPIR): The frequency of true positive alerts among recognition opportunities for individuals included in the watchlist. Note that "ground truth" about whether or not an individual in the field of view of the surveillance cameras is on the watchlist is known only for the bluelist. Consequently, the TPIR is evaluated solely from recognition opportunities of members of the bluelist.

Watchlist: List of individuals of interest to the MPS (and their associated facial images and metadata) for detection by LFR.

Zone of Recognition: 3-dimensional space within the field of view of the camera and in which the imaging conditions for robust facial recognition are met. In general, the Zone of Recognition is smaller than the field of view of the camera, e.g. not all faces in the field of view may be in focus and not every face in the field of view is imaged with the necessary resolution for facial recognition.

Acronyms

FNIR False Negative Identification Rate
FPIR False Positive Identification Rate
FR Facial Recognition
IDENT1 UK national automated fingerprint identification system
LFR Live Facial Recognition
MOPAC Mayor's Office for Policing and Crime
MPS Metropolitan Police Service
NPL National Physical Laboratory
PIA Privacy Impact Assessment
PNC Police National Computer
PTZ Pan, Tilt, Zoom
SCC Surveillance Camera Commissioner
TPIR True Positive Identification Rate

2 Executive Summary

This report provides an evaluation of the trials of public deployments of 'fixed plot' Live Facial Recognition (LFR) technology by the MPS between August 2016 and February 2019. The report goes into the details relating to the development and conduct of the trials, the lessons learnt and findings identified by the MPS.

This report concludes that Live Facial Recognition is a valuable crime-fighting tool that has the potential to help the MPS prevent and detect crime, preserve public safety and bring offenders to justice. It makes a number of recommendations for the effective and proportionate use of LFR technology.

Outline of Purpose and Trial Structure

The purpose of the operational trial was to assess the value, viability and challenges (including technological, legal, ethical, and governance) of integrating LFR technology as a policing tool to help facilitate the identification of subjects of interest in a particular location. Value can be measured in a number of ways, both in terms of the monetary costs involved in deployment, but also in the public value derived from taking a dangerous criminal off the streets who may have caused further serious harm had they not been brought to justice so soon. The latter measure is harder to quantify, except in terms of trying to assess whether a similar amount of financial investment spent in other ways could realistically achieve the same results over time.

The trial utilised a Facial Recognition (FR) system directly connected to a limited number of portable cameras. These were specifically set-up in positions within a fixed geographic location (`fixed plot') for the period of the operational deployment. As part of the deployment, MPS Officers were available to immediately review and adjudicate on alerts generated by the LFR system.

The trial involved ten operational deployments in a range of physical and environmental conditions, during which time the watchlist size was eventually increased to more than 2000 subjects. Different cameras were used, according to the footprint of the deployment. The FR algorithm was updated to the latest version available mid-way through the trials in November 2017.

Tactical Outcomes

A key measure for this trial is the outcomes generated from utilising LFR for the identification of subjects of interest, in addition to evaluating technical performance of the LFR system.

SUMMARY OF TACTICAL OUTCOMES	
Number of deployments	10
Combined duration of deployments	Approx. 69 hours
Watchlist size	Ranging from 42 to 2401
Recognition opportunities (number of people appearing video)	Approx. 180,000
Number of people engaged by a police officer following alert by the facial recognition system	27
Number of alerts confirmed correct at engagement	10
Actions / Arrests as result of alert	9

The increase in watchlist size is believed to have been a key contributor to the fact that 89% (8) of the total number of identification and arrests made during the trials occurred in the final 4 deployments. It should be noted that these arrests are directly attributable to identifications made following alerts generated by the LFR system. Additional arrests were also made as a result of proactive policing by officers attached to the LFR deployment.

Comparison with `Manhunt' Tactics

The `manhunt' tactic, where officers seek to locate a named individual for a serious offence, is a helpful comparator for benchmarking the benefits of LFR. A wide range of tactics are utilised during manhunts to locate and arrest offenders. The tactics include the deployment of officers to multiple locations for extended periods in order to identify potential locations for the offender. Many `manhunts' for offenders wanted for very serious offences such as murder involve hundreds of officer and staff hours. When aggregated together, manhunts cost many thousands of policing hours across London.

By comparison, the final four trial deployments of LFR resulted in eight arrests. So even before any anticipated improvements in these statistics as the deployments further improve, LFR can be seen to offer a favourable comparison when considering the overall resources invested in the location of wanted offenders. It should be noted that LFR can be used to complement current practices, for example operations at transport hubs, in order to improve the outcomes, increase operational efficiency and effect more arrests of offenders.

Comparison with Stop & Search Tactics

LFR deployments provide opportunities for police officers to engage with a person potentially wanted by the police and the courts. Another relevant comparative metric for LFR are the policing outcomes resulting from `stop and search'. In the past year, 13.3% of Stops in the MPS resulted in an arrest. By contrast, 30% of engagements following an adjudicated alert from the LFR system resulted in the arrest of a wanted person.

System Accuracy with respect to different demographics

The media have reported that FR systems show 'racial bias'. Meta-analysis of data from a controlled test and the trial operational deployments have demonstrated that differences in FR algorithm performance due to ethnicity are not statistically significant although differences by gender are more marked. These results have enabled the MPS to consider the adjudication process to ensure it properly responds to any variations in the generation of alerts in accordance with its Equality Act 2010 duties.

Key Findings

The MPS assess that the LFR operational trials indicate that LFR technology has provided, and will continue to provide financial and public value. The MPS believes that the technology has reached a stage where it is viable. Similarly, the ethical and legal aspects associated with its use, like many other tools, can be appropriately managed with the support of a detailed structure setting out how LFR is to be used. Robust accountability can be delivered through strong governance processes.

The trials indicate that LFR will help the MPS stop dangerous people and make London safer. Specifically it will:

- help the police to prevent and detect crime, aiding officers to identify individuals wanted by the police and courts;
- help the police to improve security and safety on the streets and at public events, particularly when helping to identify persons who pose a significant risk to the public;
- help the police to protect borders and important infrastructure where criminals and other dangerous persons may try to avoid being identified.

The value of LFR can be applied in a variety of locations.

- Notting Hill Carnival; this environment presents a number of challenges to current LFR technology due to the large volume of people and the multiple points of ingress onto the carnival footprint. The deployments here proved that careful consideration must be given to where LFR cameras are deployed and how the technology can operate effectively within a defined space.
- Remembrance Day / Port of Hull; locations with a narrow flow of people, all moving towards the cameras, provide an ideal environment for the use of LFR technology.

- Town locations such as Stratford, Soho and Romford; The location and configuration of cameras must be carefully managed in order to optimise LFR performance.

Key Recommendations

Based on the operational trials, a number of recommendations are made. An underlying principle throughout is that decisions are made by humans and not the technology. The key recommendations are:

- Locations; Should be supported by an intelligence case for deployment, and where the flow of people, crowd density, and camera performance are all suitable.
- Watchlists; Each watchlist should be created bespoke for a deployment based on the policing purpose, the potential for those on the watchlist being found at the deployment location. Necessity and proportionality must be satisfactorily articulated.
- Watchlists; Back-end automation should be used to drive efficiency, accuracy and reliability of each watchlist.
- The `Human-in-the-loop'; Decisions should be made by people. This includes deciding where and how LFR should be used, as well as each and every decision to engage a member of the public.
- Resources: Each operation must be sufficiently resourced so as to be able to respond appropriately to the alerts generated by the system.

Summary

It is possible to look back at nascent stages of the MPS LFR trial with the benefit of today's understanding of LFR and how it can best be used. Naturally, today's views are informed by the subsequent discussion as well as the detailed legal consideration given to LFR by the courts. Of course, a number of improvements to how the MPS uses the system were made during the course of the trials as part of the ongoing learning process. Indeed, the MPS's operational trial has contributed to the development of a framework for the introduction of new technologies. The Home Office Biometrics strategy [2][2] and the London Police Ethics Panel 'Final Report on LFR' [3] published in June 2018 and May 2019 respectively, set out ethical advice with regards to things that should be considered ahead of introducing new biometric applications. This ethical guidance is a welcome step forward as law enforcement seeks to make best use of technology in fighting crime.

3 Introduction

3.1 Background

It is incumbent on the MPS to ensure it explores new technology to mean it is best placed to fight crime in an increasingly complex environment. In 2009, the MPS implemented a Facial Recognition system of custody images for the purposes of identifying subjects of interest in criminal investigations. Since then, the MPS has continued to evaluate technological advances in Facial Recognition against a number of different potential use cases.

One use case for Facial Recognition is the real time identification of subjects of interest to the police. For example, at any point in time, the MPS are actively pursuing several thousand individuals who are wanted for arrest by the police or wanted on arrest warrants issued by the Courts. All of those wanted are circulated on the Police National Computer (PNC). The offences people are wanted for range in crime type and seriousness. To ensure police services reduce public risk, and to maintain public safety and confidence, the MPS aims to reduce the number of offenders 'at large' and maintain the lowest possible numbers of wanted offenders at any one time. Existing methods for locating wanted individuals can be costly and time and resource intensive.

Following successful bench evaluations of LFR using volunteer subjects, in 2016 the MPS proposed an operational trial of LFR for the real time identification of subjects of interest use case. This proposal was assessed against the House of Commons Science & Technology committee Report on the 'Current and Future uses of biometric data & technology' [4] which made a recommendation that '*rigorous testing and evaluation must therefore be undertaken prior to and after, deployment and details of performance levels published*'. The report also acknowledged that '*testing on artificial or simulated databases tells us only about the performance of a software package on that data. There is nothing in a real technology test that can validate the simulated data as a proxy for the 'real world*'. It was, therefore, very clear that the MPS should not simply roll out LFR technology, but needed to undertake an evaluation using real data in a real world context.

After due consideration, the MPS set an objective to trial LFR technology in order to understand its potential as a tool for operational policing. Previous bench tests had exhausted the level of information that could be gained in simulated conditions. In order to test if LFR technology could translate to an effective policing tactic, it was essential that an operational evaluation using real data was undertaken.

The strategic, operational and technical objectives of the trial can be summarised as:

a) To generate an evidence-base for the overt use and deployment of LFR technology as a policing tactic.
b) To ensure that all relevant legislative provisions are complied with and the overt use and deployment of LFR for policing purposes meets the oversight and regulation framework outlined in the UK by the Surveillance Camera Commissioner, the Biometrics commissioner and the Information Commissioner.
c) To build trust and confidence amongst London's communities through the overt use and deployment of LFR technology for policing purposes.
d) To ensure that societal and ethical considerations are addressed from the outset.
e) To adopt a robust, proportionate and intelligence-initiated approach in engaging individuals identified on the 'watchlist' at selected events.
f) To conduct an evaluation and provide objective evidence into the effectiveness of the overt use and deployment of LFR technology as a policing tactic that meets International Standardised methodology.

In order to fully evaluate the operational application of the technology, the intention was to run ten trial deployments over a diverse set of scenarios, varying in terms of location, watchlist of subjects of interest, throughput of individuals, environmental conditions, and policing objectives or outcomes. The National Physical Laboratory were engaged to assist with the evaluation and review of technical system performance during the trials.

The trials aimed to provide an evidence base for strategic decision making as to the potential effectiveness of LFR as a policing tool and to determine:

a) The performance that can be anticipated in operational LFR deployments in terms of the end-to-end process, including human adjudication of LFR alerts.
b) The factors that significantly influence LFR performance and those which should inform the planning and decision making process when considering a proposal to deploy LFR.
c) Identify any desirable functionality that is missing from the current facial recognition solution that would improve the system in terms of technical performance or ease of operation.

As an exploratory investigation into the effectiveness of the deployment of a new technology, there was no 'baseline' against which to compare performance. As such, an acceptance criteria, for example in terms of the number of arrests per deployment, could not be set.

Although the timescales for running the trial deployments were not specified or fixed, it took longer than expected to complete all ten: two at Notting Hill Carnival, one at the national Remembrance Sunday event, one at Port of Hull, two at Stratford Westfield, two in Soho, and two in Romford.

3.2 Legal and Governance

As an emerging technology, LFR is not subject to dedicated legislation. . However, prior to the trials, the MPS took into consideration the manner and legal basis under which the system would be used, the retention, review and deletion of data recovered, the use of the system for overt surveillance purposes as well as the ethical concerns with respect to invasions of privacy and counter arguments against such use.

Ahead of the operational trial of LFR, the MPS undertook a significant period of engagement and consultation with the offices of the Surveillance Camera, Information and Biometrics Commissioner. The MPS is are grateful to them for their input, advice & guidance, which contributed to MPS thinking around the use of such technology and assisted the MPS with the commitment to adhere to all existing and relevant policy and governance. Within the Science & Technology committee Report, the ICO stated that '*the DPA [Data Protection Act 1998], was technology-neutral and adequately flexible to ensure that biometric data can be processed in compliance with the essential legal obligations and safeguards*' and therefore the MPS welcomed, in particular, discussions with the Information Commissioner's office with respect to the Privacy Impact Assessment and updated Data Privacy Impact Assessment [5]. Likewise, the MPS completed the Surveillance Camera Commissioner's Self-Assessment Tool against the twelve guiding principles of the surveillance camera code of practice.

Prior to the first trial at Notting Hill Carnival, the MPS sought the views of community groups and the civil liberty group, Big Brother Watch. An area of concern that was highlighted was the potential of the LFR system to collect facial images from people, which might be added to a database for subsequent searching. The MPS was able to provide reassurances that this was beyond the remit of the use of LFR and indeed that there were built in 'privacy by design' features in the system that prevented such an application.

Engagement & consultation continued throughout the trials period and the MPS incorporated recommendations from documents such as the ICO 'Code of Practice for Surveillance Cameras and Personal Information'[6] published in 2017 and the RUSI report on 'Machine Learning Algorithms and Police Decision Making; Legal, Ethical & Regulatory Challenges' [7] published in September 2018.

In 2018, the MPS documented a Legal Mandate for use of LFR on a trial basis and subsequently published this document on its website [8].

The Legal Mandate identified the police's common law powers to prevent and detect crime, preserve order and bring offenders to justice as providing a robust legal power for the MPS to undertake LFR trials. Measures were also developed and taken to ensure Article 8 human rights requirements of necessity and proportionality were respected. In addition to a number of measures being designed into the LFR system, data protection, privacy and equalities impact assessments were also conducted and reviewed to inform the MPS and ensure compliance with data protection and equalities legislation.

The courts have recently considered the use of South Wales Police's trials of Facial Recognition technology in R (on the application of Edward Bridges) v The Chief Constable of South Wales [2019] EWHC 2341 (Admin). The court concluded that the police's common law powers to support the use of LFR were "amply sufficient". The court further considered the human rights points and decided that whilst Article 8 was engaged, the use of LFR was necessary and proportionate in the circumstances. The court also accepted a number of use cases, including identifying individuals unlawfully at large having escaped from custody, identifying persons with outstanding warrants for their arrest as well as other uses including protecting the public at events and helping the vulnerable. Identifying a number of important safeguards to the use of LFR including with regards to the Public Sector Equality Duty, the court dismissed the challenge on all grounds and found the use of LFR in the case to be lawful.

Over the course of the trial, the MPS has provided evidence on the use of LFR by Law Enforcement to both the Home Office Biometrics & Forensics Ethics Group and the London Policing Ethics Panel.

3.3 Objectives

Objectives relating to technical performance of the LFR system were:

- To determine the performance that can be anticipated in operational LFR deployment; and
- To identify factors that significantly influence LFR performance and to help establish guidance on configuration of LFR to optimise controllable factors for future deployments.

3.4 Concept of operations

The operational evaluation was designed to assess the end-to-end integration of LFR for the identification of subjects of interest, into a policing deployment. The LFR System deployed for the trial consisted of the NEC Neoface facial recognition software on an integrated server and client with monitor, hardwire connected to Commercial Off The Shelf (COTS) cameras. The cameras were configured and optimised specifically for each environment. Alerts on the system were transmitted over a private access point to handheld devices with the Neoface App. The end-to-end application was deployed as a closed system on a fixed plot for the period of the operational deployment. A number of 'privacy by design' features were incorporated into the LFR system and individuals who were present in the deployment area were not added to or retained on a database for subsequent processing. This is an important privacy safeguard which is often lost in the wider public debate surrounding the MPS's use of LFR.

As people walked towards the cameras and in to the Zone of Recognition, the faces detected in the footage were extracted and compared against the facial images on the watchlist. Scores above the set threshold generated an alert on both the computer running the system and on Android devices issued to officers supporting the deployment. Faces detected, but not matched, were immediately discarded by the system. An officer reviewed the alerts and undertakes an adjudication to make a decision on whether to engage with the subject. If the subject is engaged, further checks were made to confirm the identification, and appropriate action taken if required.

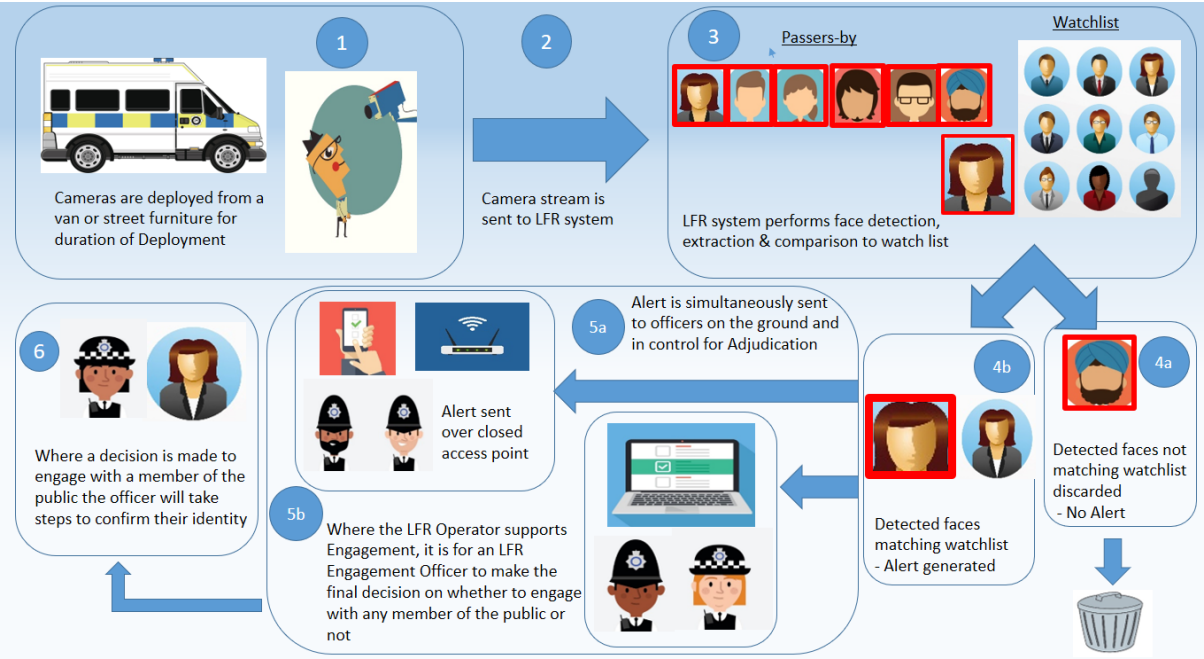


Figure 1 - Concept of operations

This concept of operations results in a filtering mechanism for dealing with face detection and alerts, which has a number of stages, as described in section 3.4.1 – 3.4.5 below.

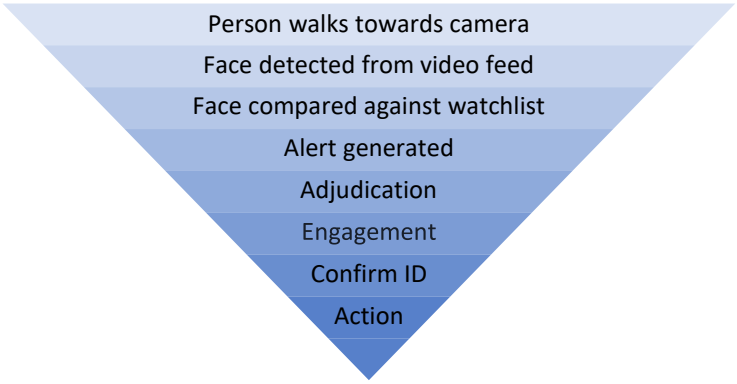


Figure 2 – The processes in filtering recognition opportunities to find a person on the watchlist

3.4.1 Subjects walk through the Zone of Recognition

The Zone of Recognition is the 3-dimensional space within the field of view of the camera where the imaging conditions for robust facial recognition are met. In general, the Zone of Recognition is smaller than the field of view of the camera, so people might appear in the video feed, but their faces might not be processed by the facial recognition system.

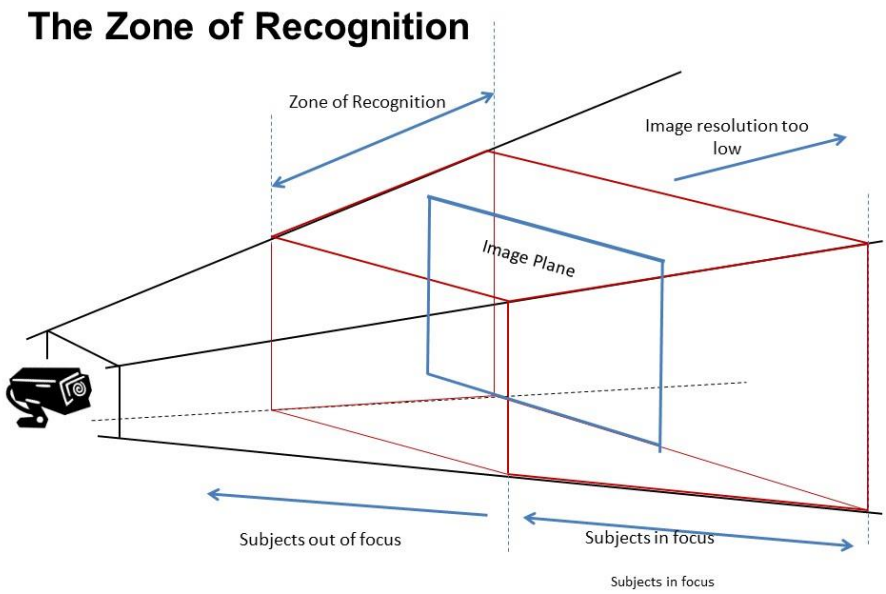


Figure 3 – Pictorial representation of the Zone of Recognition

Following feedback from the Information Commissioners Office and Civil Liberty groups, the signage advising people of the trial was updated and placed in advance of the Zone of Recognition so people could choose not to walk past the LFR cameras. Signage and leafleting about the facial recognition trial, and the police presence at each deployment, may plausibly have diverted some of the watchlist subjects away from the deployment area.

3.4.2 LFR system generates an alert when a detected face matches a watchlist face image

The LFR system analyses frames in the video feed, detecting faces and comparing these against those on the watchlist. When a comparison score exceeds the threshold, the system alerts officers to the potential match. An individual will generally appear in several video frames of the recognition opportunity, and to prevent the LFR system generating repeat alerts, further matches against the same watchlist image are suppressed for a configurable short period of time that is sufficient for the recognition opportunity to have completed.

Alerts were presented in two ways; on the LFR system's computer monitor and on mobile devices issued to officers on the ground. Officers on the ground were equipped with a mobile phone or tablet with the NeoFace Watch App installed and were stationed downstream from the cameras and LFR system so that when they received an alert on their device, the matched individual would be moving towards them allowing them to pick them out of the crowd. Officers were able to examine the match and metadata details, and to display the full frame of video which provides context such as the clothing and associates of the matched individual. All images associated with an alert were retained for thirty days and then destroyed. The exception to this is if the individual is subject to criminal justice system prosecutions, in which case the images are retained in accordance with MPS retention, removal & destruction policies reflecting MoPI and CPIA.

3.4.3 Adjudication

Officers must adjudicate alerts and decide whether to engage with an individual when an alert occurs. Adjudication can be undertaken by either the officers in front of the LFR system or by officers on the ground.

Due to the nature of Facial Recognition algorithms, the FNIR and TPIR rates and the underlying factors which can influence them (including but not limited to the environmental conditions in which the LFR system is operating), the adjudication process is an important aspect of how the LFR system is used. Adjudication ensures the use of

LFR and the engagements stemming from it remain proportionate to the purposes of the deployment. It means that officers can consider factors which may impact on the accuracy of an alert and the likelihood that the alert is incorrect as a result. Ultimately it means officers made the decision on any engagement rather than the LFR system.

3.4.4 Engagement

Officers were stationed downstream of the Zone of Recognition with sufficient distance to allow them the time to examine the alert and locate the person for engagement purposes. Officers were briefed prior to the LFR deployment and were informed that LFR provides a potential intelligence lead that must be assessed in order to instigate an engagement with an individual. The engagement held no separate legal power to detain. As such, conventional policing processes were to be followed and officers were to interact with members of the public as in the normal course of business, albeit with LFR acting as an aid to officers making an identification of a person of interest to the police.

Officers were also briefed that individuals who avoided the LFR system should not automatically by stopped. However, officers should use their discretion and judgement, as per their standard policing processes, to engage with an individual.

During an engagement, it was explained to the individual that a LFR deployment was taking place in the locality and that the system had generated an alert as they passed the system cameras. Leaflets providing information about the LFR trial, with details of an email address to contact, were provided to all individuals engaged with (see Annex D for an example of the information provided). Officers were briefed, in the first instance, to request the individual's name in order to confirm who there were.

In particularly busy environments, for example Soho, there were occasions where a decision was made to engage with an individual, but the person could not be located in the crowd. The ability to locate an individual was sometimes hampered by the alert being generated just as the individual was moving out of the field of view of the camera so that the context image only showed the persons face and not their clothing.

3.4.5 Confirm ID / Action

Methods available to confirm the identity include PNC checks, visual inspection of any identification documents offered for examination and, where available, mobile fingerprint devices. If a subject refused to provide any information, identification documentation or fingerprint, officers used their judgement against the National Decision Making Framework as to the appropriate next steps (if any) to take.

4 Trial methodology and metrics

4.1 Data collected per deployment

For each deployment, data was collected to enable measurement and reporting of LFR performance.

Performance is based on events and outcomes occurring within active duration of the deployment, starting after completion of all set up activities (i.e., camera system configuration officers ready to perform adjudication and engagement with subject alerts, and with staff on hand to record bluelist recognition opportunities), and ending just before closing down activities.

Some data is logged automatically by the LFR system, while other data must be recorded by hand, or estimated from samples of recorded video:

- Logged automatically by the LFR system

  The LFR system automatically logs all alerts. The log includes details whether the alert is against the bluelist or the operational partition of the watchlist, together with details of time, camera, comparison score, and some watchlist metadata. This information provides details on the number of alerts arising from recognition opportunities of members of the public (crowd) or bluelist.

- Recorded by hand

  The results of adjudication, engagement, confirmation of ID, and details of any action were recorded for each alert.

  Details of recognition opportunities by members of the blue list were also recorded and later reconciled with the logged system alerts for the bluelist to determine the number of missed alerts for the bluelist.

- Estimated

  The recorded video was retained for up to one month before deletion after each deployment. The video record was used to estimate crowd throughput and demographics, and to provide example footage for reporting purposes (after redaction of faces).

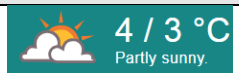
  Several short sections of the video were sampled from the active duration period of the deployment. By counting the number of people passing through the Zone of Recognition, and noting their demographics (deployments 6-10), or “face detections” as determined by the display of bounding box around the face (deployments 1-5), estimates were made of crowd throughput, crowd demographics and face detection rate for the deployments.

- Additional information recorded includes

  Camera models
  The weather at the location on the date (retrieved from the historic weather record at: www.timeanddate.com/weather/
  Sketch showing approximate layout of the deployment
  Assessed ethnicity of the operational watchlist subjects (last 5 deployments)

An example of the data collected for each deployment is shown in 4. The details, for all the deployments are provided in Annex B.

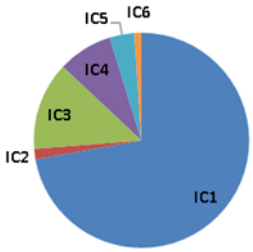
Deployment details

31 January 2019 

Duration 7 hr 10 min

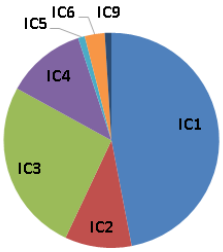
Environment: Free flowing – no control

Crowd throughput 1020 per hour (estimate)



Crowd: perceived ethnicity

Watchlist size: 2401

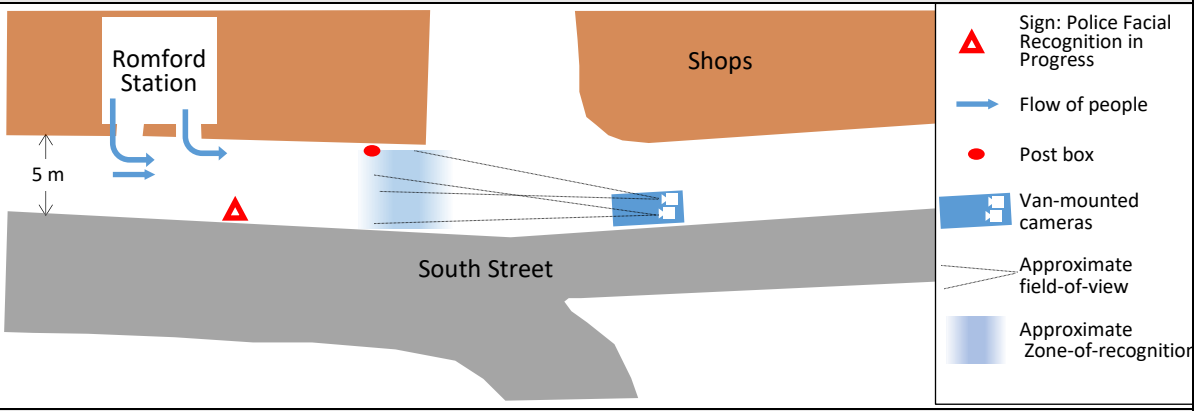


Watchlist: perceived ethnicity

Outcomes

Crowd		Bluelist	
# Recognition opportunities	7300 (estimate)	# Recognition opportunities	70
# Alerted	10	# Correctly Alerted	46
# Engaged	5		
# Alert confirmed correct	3		
# Action	2		

Layout



Example footage



Figure 4 – Example of data collected per deployment (Romford Feb 2019)

4.2 Performance metrics[1]

The evaluation has been conducted to follow the requirements and recommendations of the standards ISO/IEC 19795 on Biometric performance testing and reporting [9][10][11], and ISO/IEC 30137 Part 1 & Part 2 on the Use of biometrics in video surveillance systems [12][13].

There is no single figure that can be used to describe the 'accuracy' of a facial recognition system in any meaningful way. The standards mandate reporting performance of identification systems in terms of the frequency of **two** error conditions of the identification process; false positives and false negatives. The error rates will be measured over recognition opportunities, i.e. the period that a subject is walking through the Zone of Recognition.

The **False Positive Identification Rate** (FPIR) is the proportion of recognition opportunities of subjects who are not on the watchlist which generate an alert:

$$\text{FPIR}(N,T) = \frac{\text{Num. recognition opportunities of subjects not on the watchlist that generate an alert}}{\text{Num. recognition opportunities of subjects not on the watchlist}}$$

where N represents the size of the watchlist, and T the threshold that the comparison score must exceed for an alert to be generated.

The **False Negative Identification Rate** (FNIR) is the proportion of recognition opportunities of subjects who are on the watchlist which don't generate the correct alert.

$$\text{FNIR}(N,T) = \frac{\text{Num. recognition opportunities by subjects on the watchlist not generating a correct alert.}}{\text{Num. recognition opportunities by subjects on watchlist}}$$

FNIR states the "miss" rate. Sometimes it is preferred to talk in terms of "hit" rates. The complement of FNIR is the **True Positive Identification Rate** (TPIR).

$$\text{TPIR}(N,T) = 1 - \text{FNIR}(N,T).$$

[1] Performance results given in this report pertain to a single FR vendor and one particular model for LFR implementation. Performance for other facial recognition software may be different not least as this is an evolving technology.

4.2.1 Determination of FPIR

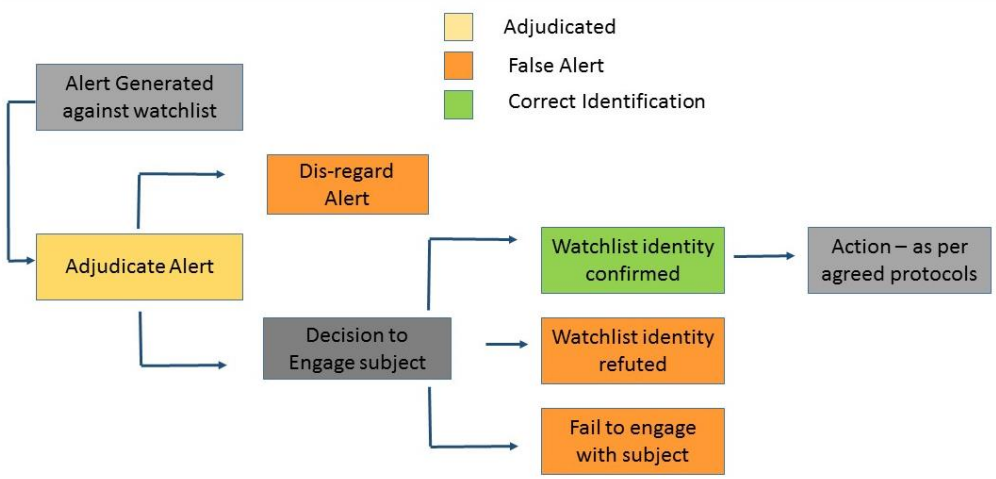


Figure 5 – True & False alerts for purpose of evaluation

In this evaluation, the MPS has included in the count of false alerts, all facial recognition alerts that were not subsequently confirmed at engagement with the individual present through identity documentation, PNC checks or via IDENT1. This may fallaciously increase the count of system False Positive Alerts, as decisions to disregard the alert and/or failure to engage with the person might actually be based on correct alerts. However, without a confirmation of identification, this is the most transparent way to count alerts.

It is worth noting that, other than for bluelist, almost all recognition opportunities are by people not on the watchlist (as the prevalence of "Wanted Missing" among the general population is very low).

Thus, (with removal of data from bluelist recognition opportunities) the False Positive Identification Rate can be estimated as:

$$\text{FPIR} \approx \frac{\text{Number of alerts} - \text{Number of confirmed identifications}}{\text{Number of recognition opportunities}}$$

The number of recognition opportunities is estimated from analysis of several short samples of video footage, as described in section 4.1.

4.2.2 Determination of TPIR

Determination of the True Positive Identification Rate is made based on recognition opportunities by **bluelist subjects only**, as the trial has no way to count the number of people on the operational watchlist that are missed by the LFR.

$$\text{TPIR} = \frac{\text{Number of correct bluelist alerts}}{\text{Number of bluelist recognition opportunities}}$$

It should be noted that images of bluelist subjects are seeded into the full watchlist and that bluelist subjects are compared against the totality of the watchlist, not just the blue list partition.

5 Deployment Outcomes

5.1 Summary of deployments

Across the ten deployments a number of different factors were evaluated. The majority of the deployments were outdoors with a free flow of subjects towards the cameras, which were either mounted on street furniture or on a van and set up specifically for the duration of the deployment. Although free flowing, there were differences in the field of view from narrow to wide. The watchlists primarily comprised individuals who were 'Wanted Missing' for a range of different offences, dependent on the specific operational imperative. The watchlist size increased from circa 250 at the first deployments to over 2000 individuals on the last four deployments.

The Remembrance Day deployment was outdoors but differed in that there was a controlled queue of people and the watchlist consisted of individuals whose presence was likely to compromise the security or safety of the event.

The Port of Hull deployment took place indoors and the LFR system was integrated into existing camera infrastructure.

5.2 Notting Hill Carnival, 28-29 August 2016

The purpose of this first deployment was to test the end-to-end integration of the technology into an operational policing deployment and build practices around rapid deployment & creating a watchlist. Initially the intention was to deploy cameras to a transport hub, where there would be a level of control over the flow of people through the barriers. However, due to circumstances outside the control of the MPS, the LFR technology was deployed from the ground with cameras on a 'boom' extended over a street. This was challenging with respect to the width of the ingress point, and the lack of control over the crowd flow. The criteria for inclusion on the watchlist was aligned with key crime areas being targeted including wanted offenders for sexual offences, 'theft person' and individuals on bail with specific conditions not to attend Notting Hill Carnival.

Notting Hill Carnival 2016 - Summary	
Duration	12 hours
Watchlist size	266
Recognition opportunities	Approx. 15,900
Alerts against operational watchlist	1
People engaged by a police officer following alert	0
Arrests / actions	N/A
False positive identification rate	*0.01%*
True positive identification rate for Bluelist	*54%*

Although there were no positive identifications against the watchlist, the algorithm performance under such challenging conditions, combined with the use of the technology by officers, demonstrated the potential of LFR.

5.3 Notting Hill Carnival, 27-28 August 2017

The purpose of this deployment was to build on the lessons learned in the first deployment, and to test the use of 360º PTZ cameras deployed from a vehicle. The environment represented an uncontrolled flow of a high density of subjects with a wider area of coverage than the previous deployment. The watchlist criteria were again aligned with the same key crime areas being targeted as for Notting Hill Carnival 2016 and almost doubled in size.

Notting Hill Carnival 2017 - Summary	
Duration	12 hours
Watchlist size	528
Recognition opportunities	Approx. 101,000
Alerts against operational watchlist	96
People engaged by a police officer following alert	6
Alerts confirmed correct at engagement	1
False positive identification rate	*0.09%*
True positive identification rate for Bluelist	*71%*

It may initially appear that 95 is a large number of false alerts. However, this must be considered in context of the number of recognition opportunities, which exceeded 100,000 resulting in a FPIR of 0.09%. Exploring false alerts further revealed that a significant proportion (almost 50%) were generated due to similarities in pose, illumination or expression of a subject watchlist image when compared to the facial image captured of individuals by the LFR system.

One of the individuals that was stopped and had his identity confirmed through PNC checks. No further action was taken on the basis of this identification as the individual had been dealt with the previous week. This subject was still on the watchlist because at the time, the process to create watchlists was significant and lengthy. This

trial identified the need for the process to be overhauled to ensure that watchlists can be produced sufficiently quickly to ensure their reasonable currency.

The TPIR of 71% (for Bluelist subjects) provided assurance that the technology was capable of generating alerts against individuals present and on the watchlist.

5.4 Remembrance Sunday 12 November 2017

This deployment represented a controlled flow of people though the use of 'tensator' barriers, such that there was only 3 – 4 faces in the 'Zone of Recognition' at any one time. The LFR systems were deployed at two sites to cover all ingress points into a secured area. The NeoFace Algorithm was updated from S17 to M20 on one of the systems. The watchlist criteria consisted of individuals whose attendance would pose a risk to the security and safety of the event.

Remembrance Sunday 2017 - Summary	
Duration	3 hours 30 minutes
Watchlist size	42
Recognition opportunities	Approx. 12,800
Alerts against operational watchlist	7
People engaged by a police officer following alert	2
Alerts confirmed correct at engagement	1
Actions	1
False positive identification rate	*0.05%*
True positive identification rate for Bluelist	*89%*

The subject whose identity was confirmed was unable to gain access to the secure area but no arrest was deemed necessary in the circumstances. The controlled nature of the flow of people, combined with the camera siting and configuration resulted in the highest TPIR of all the deployments and demonstrates the value of the LFR system for the police to discharge its responsibilities in a public safety context.

5.5 Port of Hull, 13-14 June 2018

This deployment tested two different aspects; an indoor environment (with a level of control of subjects towards the cameras) and the ability to integrate the LFR capability into existing CCTV infrastructure. From this deployment onward, the NeoFace M20 algorithm was used. The watchlist criteria (set by Humberside Constabulary) was constrained to subjects wanted for criminal offences based on the current crime analysis and priorities.

Port of Hull 2018 - Summary	
Duration	5 hours
Watchlist size	144
Recognition opportunities	Approx. 800
Number of alerts	0
False positive identification rate	*0.0%*
True positive identification rate for Bluelist	*80%*

Although there were no positive identifications or arrests, the trial met its objective and demonstrated that LFR can be integrated into the existing CCTV infrastructure and that rapid deployment of LFR can provide additional capability. For example, at a smaller or remote port, where a Facial Recognition (FR) system is not required at all times, an LFR system may be needed to respond to a particular threat or use case where its deployment would meet a necessity threshold and could be proportionate in the circumstances.

The false alert rate was 0%, which might be attributed to a number of factors such as the relatively small number of people captured during the embarkation and disembarkation of the ferry, and the demographic of the watchlist being very different to the demographic of the ferry passengers.

5.6 Stratford Westfield 28 June 2018 & 26 July 2018

These deployments tested the use of LFR in conjunction with other policing tactics and operations such as Operation Sceptre [14]. The watchlist comprised all 'Wanted Missing' individuals and filtered based on geographic area (proximity to Westfield Stratford). The cameras were mounted on street furniture for the duration of the deployment and then decommissioned.

Stratford Westfield June 2018	
Duration	6 hours
Watchlist size	489
Recognition opportunities	Approx. 10,000
Alerts against operational watchlist	5
People engaged by a police officer following alert	1
Alerts confirmed correct at engagement	0
False positive identification rate	*0.05%*
True positive identification rate for Bluelist	*81%*

Stratford Westfield July 2018	
Duration	6 hours
Watchlist size	306
Recognition opportunities	Approx. 12,200
Alerts against operational watchlist	1
People engaged by a police officer following alert	1
Alerts confirmed correct at engagement	0
False positive identification rate	*0.01%*
True positive identification rate for Bluelist	*73%*

This trial demonstrated that a high TPIR could be achieved with careful positioning of cameras even without a narrow controlled flow of people. There were no positive identifications made against subjects of interest on the watchlist, which could be attributed to the small watchlist size.

5.7 Soho 17 & 18 December 2018

These deployments used cameras mounted on a van and were deployed in an open environment, such that there was no natural flow of people towards the Zone of Recognition. The location was selected based on crime analysis and intelligence. The main difference in this deployment was the use of an increased size of the watchlist, which comprised individuals wanted for violent offences. Because the location was in central London, the watchlist was not filtered to any one specific geographic area.

Soho 17 December 2018	
Duration	5 hours 45 minutes
Watchlist size	2226
Recognition opportunities	Approx. 4100
Alerts against operational watchlist	5
People engaged by a police officer following alert	3
Alerts confirmed correct at engagement	1
Arrests/Actions	2
False positive identification rate	*0.10%*
True positive identification rate for Bluelist	*74%*

Due to the low footfall at Cambridge Circus site on the morning of 17 December, the location was moved to Cranbourn Street for the afternoon and remained at that location for the 18 December trial.

Of the three individuals engaged with, one was confirmed as a correct positive identification following PNC checks and was arrested for rape. One of the other individuals engaged with was not the watchlist subject. However, PNC checks revealed that the individual was nevertheless wanted and was subsequently arrested.

Soho 18 December 2018 - Summary	
Duration	5 hours 35 minutes
Watchlist size	2226
Recognition opportunities	Approx. 8,400
Alerts against operational watchlist	9
People engaged by a police officer following alert	1
Alerts confirmed correct at engagement	1
Arrests/Actions	1
False positive identification rate	*0.10%*
True positive identification rate for Bluelist	*78%*
***Note:** In two further cases the adjudication decision for an alert was to engage with the subject, but the subject could not be located by the engagement team due to crowd density.*	

The trial showed the ability to successfully identify those wanted by the police and the increase in the size of the watchlist had a direct impact on the number of arrests made.

5.8 Romford 31 January 2019 & 14 February 2019

These deployments built on the lessons learned from the Soho deployments with respect to the watchlist size. The deployment utilised cameras mounted on a van. The watchlist comprised individuals wanted for violent offences and filtered by geographic area with respect to proximity to Romford.

Romford January 2019 - Summary	
Duration	7 hours 10 minutes
Watchlist size	2401
Recognition opportunities	Approx. 7300
Alerts against operational watchlist	10
People engaged by a police officer following alert	5
Alerts confirmed correct at engagement	3
Arrests/Actions	2
False positive identification rate	*0.10%*
True positive identification rate for Bluelist	*66%*

One of the individuals engaged had his identity confirmed through PNC checks, but no further action was taken as the individual had been dealt with in the gap between generating the watchlist and running the deployment. The same individual also passed through the surveillance system later in the day, generating a second true positive alert. The adjudication process prevented a further engagement and is an example of the benefits of the person-in-the-middle control mechanism to ensure an officer rather than the LFR system makes the engagement decision. The second alert and confirmed identification has therefore been disregarded in the summary given.

The remaining two subjects who had their identity confirmed during the engagement process were arrested.

Romford February 2019 Summary	
Duration	6 hours 45 minutes
Watchlist size	1996
Recognition opportunities	Approx. 10,100
Alerts against operational watchlist	16
People engaged by a police officer following alert	8
Alerts confirmed correct at engagement	3
Arrests/Actions	3
False positive identification rate	*0.13%*
True positive identification rate for Bluelist	*69%*
Notes: *In three recognition opportunities the individual present generated two alerts. The additional three alerts are disregarded in the counts given, and in subsequent analyses.* *In a further four cases the adjudication decision for an alert was to engage with the subject, but the subject was lost to the engagement team.*	

Three individuals were confirmed as watchlist subjects and arrested.

6 Key Learning

6.1 Watchlist generation

There are a number of factors to be considered regarding Watchlist generation:

- **Governance and criteria for inclusion on a Watchlist**

  Watchlists created for each deployment were specific to the operation and/or a defined policing objective and remained active only for the duration of the deployment. Before each deployment the criteria for inclusion were set by the senior officer in charge. With the exception of the Remembrance Day deployment, all watchlists consisted of individuals who were Wanted Missing for a recordable offence. The latter four deployments (at Soho and Romford) were restricted to those wanted for violent offences. It is important that the watchlists are operationally relevant to the location or event, as it must be plausible that the subjects on the watchlist might be present in the LFR deployment area or that the inclusion is otherwise deemed necessary in the circumstances.

- **Facial image quality**

  The performance of the automated LFR system, as well as that of officer adjudication of alerts, is dependent on the quality of the images in the watchlist. In the main, full frontal face images, taken against a plain background, were sourced from an MPS custody image. Nevertheless, there were always a small proportion of images that failed to enrol in the LFR system due to image quality, or generated a lot of alerts due to lighting or pose issues.

  Best practice is to use custody images that conform to the NPIA 'Police Standard for Still Digital Image Capture and Data Interchange of Facial/Mugshot and Scar, Mark & Tattoo Images' [15]. If multiple images of the subject are available, consideration should be given to including these in the watchlist to improve the likelihood of a match. Where it is necessary to use a lower quality image, officers undertaking the adjudication process need to be aware of the risk of increased likelihood of alerts to respond accordingly.

- **Processes for creating the watchlist of facial images and associated metadata**

  The MPS, like many other forces, is, at the time of the trials, working with legacy systems and generating the watchlist is a time-consuming task as multiple independent datasets need to be consulted and cross referenced. A fully automated bulk download of custody images is not currently feasible, and a number of custody images must be down loaded manually in order to retrieve the custody record. For the purposes of piloting LFR, it was not justifiable to heavily invest in making changes to Back-end systems. However, it was established during the operational trial that this is a key factor to the generation of sufficiently large watchlists to yield results. If the use of LFR technology is to be taken forward, then there must be investment to allow watchlist selection criteria to be applied and the associated (best quality) watchlist images to be extracted. This investment would also facilitate review of the watchlist before each deployment to ensure that all the images still meet the necessity criteria for inclusion.

6.2 Camera installation / set-up

A controlled and managed flow of individuals towards the camera demonstrated advantages over a free flow of individuals, as evidenced in the TPIR rates measured at the Remembrance Day and Port of Hull deployments. However, where there is free flow of individuals there is a balance between gaining wider coverage of an area (through higher resolution cameras and/or a wide field of view) and having sufficient resolution (number of pixels) across a face for better detection and recognition.

Either 2 or 5 Megapixel cameras were used for the trials, which give an effective horizontal coverage for facial recognition of between 1.5 and 3.4m per camera. To aid camera setup on a deployment site, it would be useful for an LFR system to be able to support the operator in adjusting field of view, lens zoom, and aperture settings

to achieve optimal parameters in the desired Zone of Recognition (for example feedback on the inter-eye distance in pixels, and whether this meets recommendations).

The NeoFace system utilised for the trial was the Rapid Deployment Alien Ware laptop configuration that can support two concurrent camera streams. It was observed that when the crowd density increased the real time processing could start to 'lag'. Although higher resolution cameras are available, the hardware configuration is not sufficient to continuously process the stream (detect faces, create a template and search against the watchlist) in near real time. With advances in processing technology, it is expected that this issue would be resolved.

6.3 Algorithm configuration

The NeoFace system allows the user to adjust a number of parameters including:

- the maximum number of faces to detect per frame;
- the decision threshold score for recognition; and
- the period of time between repeat alerts on the same individual.

For all deployments the system was configured to detect a maximum of 5 faces per frame. Non-operational system tests had established that adjusting this number any higher would result in the processing servers reaching maximum capacity and system lag.

Adjusting the decision threshold downwards may increase the TPIR but can have the effect of increasing the number of alerts that must be dealt with. Likewise, adjusting the threshold upwards can mean that subjects might walk past the camera but not generate an alert. For operational deployments, the threshold used was the default recommended by the manufacturer and this proved effective at maintaining a manageable level of false positive alerts against a sufficient True Positive Identification Rate.

When there is an alert between an individual and a watchlist subject, additional or repeated alerts for that watchlist image are supressed for a configurable period of time to allow the individual to clear the Zone of Recognition. This means that the comparison score logged for each alert is that of the first recognition opportunity that scores above threshold. This may not be the highest comparison score possible for the recognition opportunity and as a consequence, the effect of using a higher decision threshold cannot be inferred from the logged comparison scores.

The decision threshold score for recognition can be adjusted per watchlist or per watchlist subject image. The advantage of the latter feature is that the threshold can be adjusted upwards for watchlist subject images that generate a high number of alerts (for example due to Pose, Illumination or Expression factors) as observed at Notting Hill Carnival in 2017.

6.4 Facial recognition performance

FPIR and TPIR (the False-Positive and True-Positive Identification Rates) for each deployment are shown in Table 1. For the later deployments, with a watchlist size of around 2000, TPIR for the Bluelist is around 70%, and FPIR close to 0.1%. It should be noted that the environmental conditions, and camera parameters (e.g. face size for subjects in Zone of Recognition) were fairly similar in these deployments, and that the performance levels achieved will differ in other conditions.

Deployment	**Algorithm Version**	**Watchlist Size**	**FPIR**	**TPIR**
Notting Hill 2016	S17	266	0.01%	54%
Notting Hill 2017	S17	528	0.09%	71%
Remembrance Sunday 2017	S17/M20	42	0.05%	89%
Hull June 2018	M20	144	0.00%	80%
Stratford Jun 2018	M20	489	0.05%	81%
Stratford Jul 2018	M20	306	0.01%	73%
Soho 17 Dec 2018	M20	2226	0.10%	74%
Soho 18 Dec 2018	M20	2226	0.10%	78%

Romford Jan 2019	M20	2401	0.10%	66%
Romford Feb 2019	M20	1996	0.13%	69%

Table 1 – False Positive Identification Rate and True Positive Identification Rate by deployment

Both FPIR and TPIR are acknowledged to be dependent on the number of watchlist images enrolled into the system. However, in the trial deployments increasing the watchlist from the 100s to the 1000s did not noticeably worsen the performance results. Increasing the watchlist size (providing all the watchlist subjects remain specific to the deployment location) should proportionately increase the number of subjects of interest found through LFR. To be able to do so without impacting TPIR and FPIR performance is clearly beneficial.

6.5 Operator adjudication

Adjudicating alerts is challenging for a number of reasons:

- The adjudication period is strictly time limited if the subject is to be engaged;
- The LFR system will generate an alert on the first facial image that matches above threshold (which may not be the clearest image);
- There will typically be an age difference between the subject watchlist image and the individual present;
- Generally main features of the matched subject and watchlist images are quite similar (otherwise the LFR system would not have generated a high comparison score).

Due to the nature of the operational trial deployments, it was not possible to attach the same group of officers to every deployment and provide them with specific training in the adjudication process but such training has been identified as important for future operational deployments.

Officers in front of the LFR system have to adjudicate based on the single alert image, whereas officers on the ground have this image, but arguably more information as they can see the subject walking towards them. There is evidence to suggest that decision making is improved when there is more information.

Currently, the metadata associated with the watchlist subject images does not include the date that the image was taken. Officers reported that having that information would assist them in the adjudication process.

6.6 Engagement with subject

There were 6 cases where engagement did not happen following an adjudicated alert. Factors that complicate the engagement process include:

- If the face alert occurs late in the recognition opportunity when subject is departing the Zone of Recognition, the context image (showing the full camera view for subject's face image) does not show detail such as clothing that can be used to help an officer locate and engage with the subject.
- Sometimes officers may be busy processing an arrest from a previous engagement.

7 The effect of subject demographics

Several studies (e.g.,[16][17]) report that performance of facial recognition is sensitive to demographics such as ethnicity, gender and age. The results of these scientific studies have been characterised in media reports as, for example, 'racial bias' but this is somewhat misleading as the extent of differences in performance vary by algorithm, and by application and these differences are observed with respect to both FPIR and TPIR. Further studies (e.g.,[18][19][20]) report that, in an operational context, differences in performance can be as much to do with image quality issues, as the underlying demographics of the subjects. These concerns lead the MPS to consider the potential demographic effects on performance within the MPS LFR evaluation and undertake a (non-operational) trial, specifically to measure the effect of subject demographics on system performance.

Demographic differences in performance of the facial recognition algorithm and can be summarised as:

- Is the FPIR higher for one demographic group than another, resulting in members of that demographic disproportionately generating (false) alerts when they are not on the watchlist?
- Is the FNIR higher for one demographic group than another, resulting in members of that demographic being less likely generate an alert, even when they are on the watchlist?

7.1 Subject demographics and FPIR

Demographic details (IC codes) of individuals present in the area of the LFR trial were recorded for the 6th to the 10th deployments. This allows an estimation of the FPIR by demographic (IC code or Gender) as shown in Table 2. Statistical analyses[2] show that the observed differences in FPIR by perceived ethnicity are not statistically significant. However, the gender difference in FPIR is statistically significant with significantly more false alerts for men than for women.

Demographic class	Estimated recognition opportunities by class	False alerts by class	Observed FPIR for class	Comment on statistical significance
IC1	24,800	23	0.09%	Differences are not statistically significant
IC2	2,300	2	0.09%	
IC3	6,100	5	0.08%	
IC4	6,700	2	0.03%	
IC5	2,900	0	0.00%	
IC6	400	0	0.00%	
Female	21,550	10	0.05%	Difference is statistically significant
Male	21,650	22	0.10%	
Total	43,200	32	0.07%	

Table 2 – False Positive Identification Rate by demographic

7.2 Subject demographics and TPIR

The operational trial deployments used a small bluelist, comprising only staff involved in the setting up and running of the trial. A proposal was made to expand the bluelist in terms of size and ethnic diversity to provide the additional data needed to help in consideration of the effect of subject demographics on system performance.

The initial intention had been to use an expanded watchlist within one or more of the remaining operational deployments, where bluelist subjects would join the flow of the public to be representative of the operational use case and the other trial parameters. It proved impossible to coordinate availability of a suitable bluelist

[2] For analysing whether differences in observed FPIR between different demographic classes are statistically significant a chi-squared test for homogeneity of proportions was used. In this report, the MPS has used a significance level of 0.1 for reporting that a difference in performance is statistically significant.

cohort for any of the remaining trial deployments and consequently an experiment using an expanded bluelist was run separately on a Metropolitan Police premises in April 2019.

The larger bluelist comprised 75 police personnel, with demographics as shown in Table 3. Custody style images of bluelist subjects were seeded into an operationally representative watchlist of 2000 custody images. There was no crowd flow of the general public as with the operational deployments, though a few (< 10) unenrolled individuals walked together with the bluelist subjects. As a consequence, it was not possible to undertake any testing on operator adjudication as every alert generated was for an individual present.

Similar to the Romford deployments, two van mounted cameras were used, which were set to cover a coned-off pathway on the parade ground.

	Female	Male	Total
IC1	20	30	50
IC2	2	1	3
IC3	3	8	11
IC4	3	8	11
Total	28	47	75

Table 3 – Expanded bluelist demographics

The bluelist provided a total of 700 recognition opportunities (8 – 10 recognition opportunities each). Table 4 shows the differences between the TPIR for different demographic groups.

Demographic class	**Bluelist subjects**	**Recognition opportunities by class**	**True alerts**	**Average TPIR for class subjects (See note)**	
IC1	50	456	230	50.9%	Differences are not statistically significant
IC2	3	30	12	40.0%	
IC3	11	106	42	40.0%	
IC4	11	108	57	52.5%	
Female	28	258	87	34.3%	Difference is statistically significant
Male	47	442	254	57.9%	
Total	75	700	341	49.1%	
Note: Because the number of recognition opportunities per bluelist subject varied, the average TPIR for each demographic class is not identical to the ratio of total true alerts to recognition opportunities.					

Table 4 – True Positive Identification Rate by demographic

Overall, a lower TPIR was measured than for other operational deployments. The weather, light drizzle, was not as clement at previous deployments and was a possible contributing factor to this. Statistical analysis[3] shows TPIR for women to be significantly lower than that for men, but differences in TPIR for different ethnic groups were not statistically significant.

There were no False Alerts generated during this extended Bluelist trial.

7.3 Summary of findings on demographic differences in performance

The findings from the extended Bluelist TPIR trial, together with the FPIR rates observed for individuals present at the last five operational deployments, demonstrate that whilst (in line with other research papers) differences

[3] For analysing demographic differences in TPIR, as each subject has multiple detection opportunities, a different statistical approach is used. Firstly, the TPIR of each individual is calculated, being the proportion of their detection opportunities where an alert was recorded. Then a "t-test" is used to assess whether the difference in the class means are statistically significant (at a 0.1 significance level).

in both FPIR & TPIR are observed with respect to ethnicity, for this algorithm and application, those differences are not statistically significant. Differences in FPIR & TPIR recognition rates with respect to gender are statistically significant, with females having a lower FPIR, and lower TPIR than males. The results relate to a single test that showed that that LFR algorithm is slightly less likely to trigger alerts in relation to women who pass an LFR camera. The large scale tests undertaken by the National Institute of Standards & Technology [21], provide reassurance around the performance of the particular algorithm tested with respect to different demographics. Awareness of algorithm performance should be included in LFR operator training to ensure appropriate consideration is given to the adjudication of every alert.

8 Recommendations

In order to move LFR from a trial to a deployable tool to aid policing, the following recommendations are made:

Location for LFR deployment:

- The location should be selected based on a specific objective and be intelligence-led.
- Ideally, there would be a level of control over the flow and density of the crowd.
- There should be a planned siting and configuration of cameras to optimise performance.

Watchlists

- There must be an operationally relevant watchlist with a decision to include an image on a watchlist being taken on the basis of necessity and proportionality to the purposes of the deployment. The number of watchlist subjects likely to be found by LFR is in proportion to the watchlist size. There is no adverse impact on the LFR system's performance with a watchlist of 2000 compared to a smaller sized watchlist.
- Back-end systems should be updated to improve the processes, timescales and resource implications around watchlist generation.
- Back-end systems should be updated to ensure that watchlists are automatically updated, for example when a subject is no longer wanted on warrant
- Watchlist image quality should where possible comply with standards, such as the NPIA 'Police Standard for Still Digital Image Capture and Data Interchange of facial/Mugshot and Scar, Mark & Tattoo Images' (full frontal face, neutral expression, uniform lighting and plain background) with the inclusion of images outside of the NPIA standard being made in cases of necessity.
- Metadata associated with watchlist images could include the date that the image was taken as this will assist in the adjudication process.

Adjudication

- The adjudication process is an important control mechanism to ensure that an officer makes the decision on an identification and any subsequent engagement rather than the LFR system itself. This process allows the officer an opportunity to identify any errors, consider any risk of bias, assess the needs to make an intervention and take any further action in a proportionate way.
- The adjudication process could be assisted by a feature of the LFR system whereby, when an alert is generated, the operator is able to replay the video relating to the alert. This would provide additional imagery and information on which to base the adjudication.
- Further academic research is required to understand factors that impact on the adjudication process, particularly around whether adjudicating with the individual present (by officers on the ground) provides better decision making than from a single image.

Resources

- The operation must be sufficiently resourced by police officers in order to respond to alerts.

9 Conclusion

As set out at the start of this report, the purpose of the operational trial of LFR technology was to assess the value, viability and challenges (including technological, legal, ethical, and governance) of integrating LFR technology as a policing tool that helps facilitate the identification of subjects of interest in a particular location.

Although the trial took longer than anticipated, this did allow for a high level of engagement with governing bodies and civil liberty groups, with time to incorporate feedback and `lessons learned' into latter deployments.

Overall, the trial indicates that the technology is effective, with an average True Positive Identification Rate of 70% and a maximum False Positive Identification Rate of approximately 1 in 1000. The latter deployments in particular, met the operational imperative of identifying and arresting offenders wanted for violent offences. The level of policing resource versus the number of arrest outcomes is positive with 30% of engagements following an FR alert, resulting in an arrest. Additional arrests were made at each deployment due to proactive policing by the officers attached to the deployment.

The trial included an evaluation of demographic effects on FR system performance (often referred to in the media as 'racial bias'). Differences in both FPIR & TPIR were observed with respect to ethnicity, but these differences are not statistically significant. Differences with respect to gender were identified in relation to the algorithm trialled. Females having a lower FPIR, and lower TPIR than males. In addition to ensuring that a highly accurate algorithm that responds in a uniform manner to different demographics is deployed, every alert is adjudicated and LFR operator training provides awareness of any potential differences.

The trial indicates that LFR will be an effective policing tool that helps the MPS stop dangerous offenders and make London a safer place.

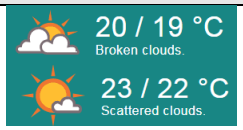
Annex A Algorithm & Camera details

Deployment		Neoface Version	Threshold	Cameras
1	Notting Hill Carnival 2016	S17	0.55	2 x Axis Q1635 – 2 Mpixel – 9-80mm lens
2	Notting Hill Carnival 2017	S17	0.55	Axis Q1635 – 2 Mpixel – 9-80mm lens Axis, P1367E – 5 Mpixel – 9-40mm lens
3	Remembrance Sunday 2017	S17 N20	0.55	Axis P1357 – 5 Mpixel – 2.8-8.5 mm lens 2xAxis Q1635 – 2 Mpixel – 9-80mm lens Axis P1345LE– 2 Mpixel – 22mm lens
4	Port of Hull 2018	M20	0.55	Port CCTV Infrastructure
5	Stratford Westfield Jun 2018	M20	0.55	2 x Axis Q1645 – 2 Mpixel – 8-80mm lens
6	Stratford Westfield Jul 2018	M20	0.55	2 x Axis Q1647 – 4 Mpixel – 8-80mm lens
7	Soho 17 Dec 2018	M20	0.55	Axis LE 8685 – 2Mpixel – 4.3-129mm lens Bosch MIC-7230-B5
8	Soho 18 Dec 2018	M20	0.55	
9	Romford Jan 2019	M20	0.55	2 x Axis Q8685 – 2Mpixel – 4.3-129mm lens
10	Romford Feb 2019	M20	0.55	

Table 5 – Algorithm and camera details

Annex B Details and example footage from each deployment

B.1 Notting Hill Carnival, 28-29 August 2016

Deployment Details			
28 August 2016		Environment	Free flowing – no control
29 August 2016		Total Duration	12 hrs 00 min
Crowd throughput	1330 per hour (estimate)	Watchlist size	266

Outcomes			
Crowd		**Bluelist**	
# Recognition opportunities	15900 (estimate)	# Recognition opportunities	56
# Alerted	1	# Correctly Alerted	30
# Engaged	0		
# ID confirmed	-		
# Action	-		

Table 6 Notting Hill Carnival 2016: Deployment detail and LFR Outcomes

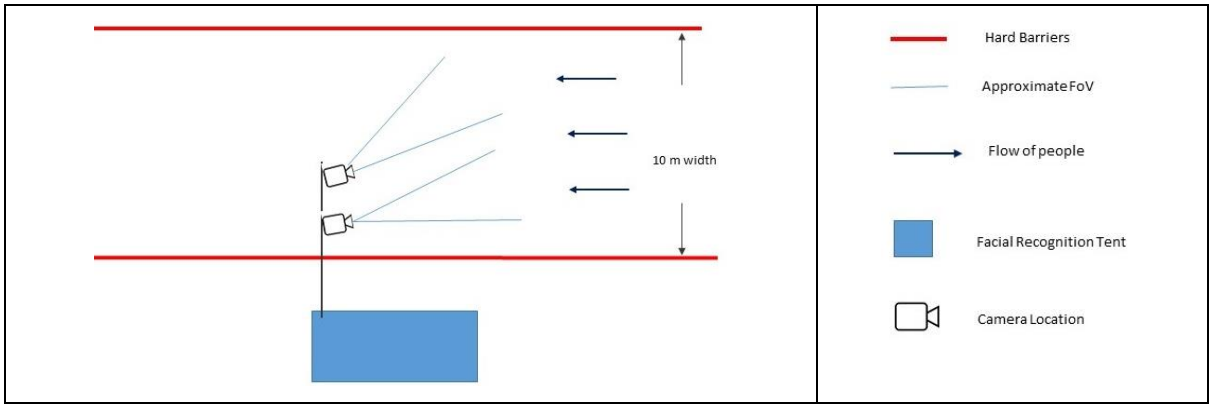
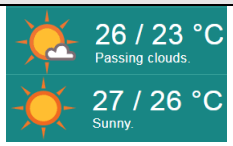


Figure 6 — Notting Hill Carnival 2016: Layout



Figure 7 — Notting Hill Carnival 2016: Example Footage

B.2 Notting Hill Carnival, 27-28 August 2017

Deployment Details			
27 August 2017		Environment	Free flowing – no control
28 August 2017		Total duration	12 hrs 00 min
Crowd throughput	8440 per hour (estimate)	Watchlist size	528

Outcomes			
Crowd		**Bluelist**	
# Recognition opportunities	101000(estimate)	# Recognition opportunities	59
# Alerted	96	# Correctly Alerted	42
# Engaged	6		
# ID confirmed	1		
# Action	0		

Table 7 Notting Hill Carnival 2017: Deployment details and LFR outcomes

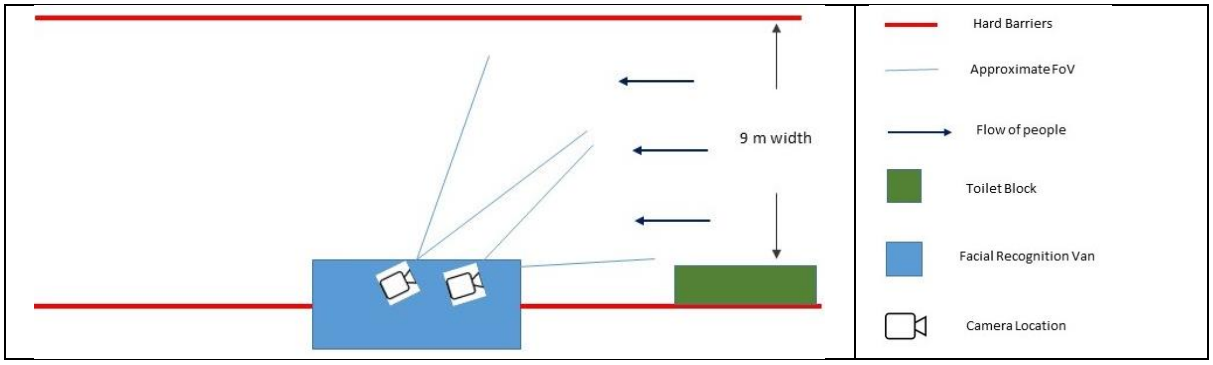


Figure 8 — Notting Hill Carnival 2017: Layout



Figure 9 — Notting Hill Carnival 2017: Example Footage

B.3 Remembrance Sunday 12 November 2017

Deployment details			
12 November 2017	7 / 6 °C Scattered clouds.	Environment	Queue controlled by street furniture and barriers
Duration	3 hrs 30 min	Watchlist size	42
Crowd throughput *Parliament Square* 2450 per hour (estimate)			
Crowd throughput *Whitehall Place* 1200 per hour (estimate)			

Outcomes			
Crowd		**Bluelist**	
# Recognition opportunities	12800 (estimated)	# Recognition opportunities	18
# Alerted	7	# Correctly Alerted	16
# Engaged	2		
# Alert confirmed correct	1		
# Action	1		

Table 8 – Remembrance Sunday 2017: Deployment details & LFR outcomes

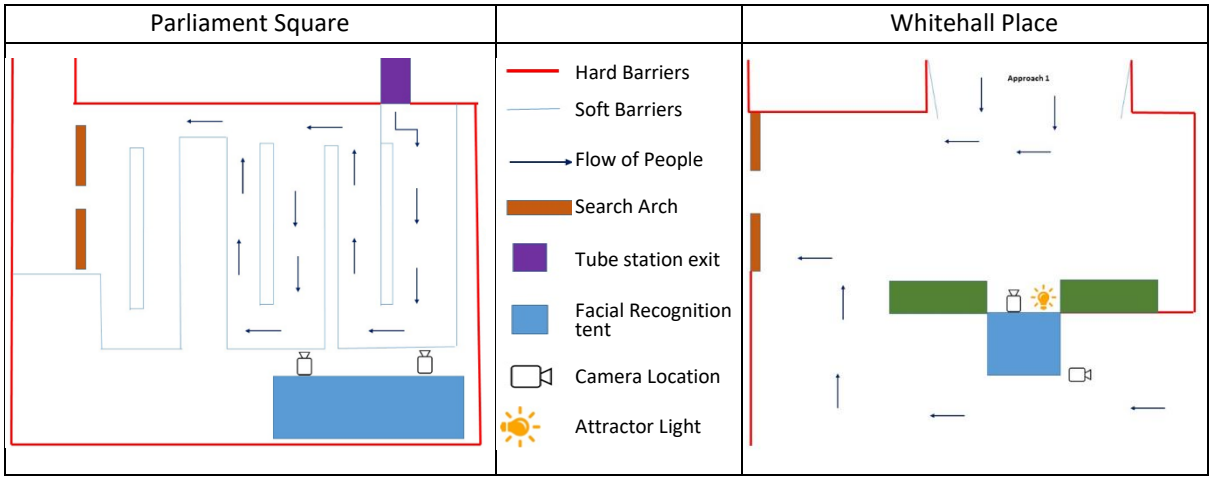


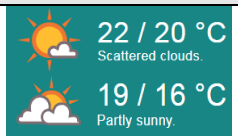
Figure 10 — Remembrance Sunday 2017: Layout



Parliament Square | *Whitehall place – Approach 1*

Figure 11 — Remembrance Sunday 2017: Example footage

B.4 Port of Hull, 13&14 June 2018

Deployment details			
13 June 2018		Environment	Internal Semi-controlled
14 June 2018		Total duration	5 hours 00 min
Crowd throughput	160 per hour (estimate)	Watchlist size	144

Outcomes			
Crowd		**Bluelist**	
# Recognition opportunities	800 (estimated)	# Recognition opportunities	66
# Alerted	0	# Correctly alerted	53
# Engaged	-		
# Alert confirmed correct	-		
# Action	-		

Table 9 – Port of Hull 2018: Deployment details and FR Outcomes

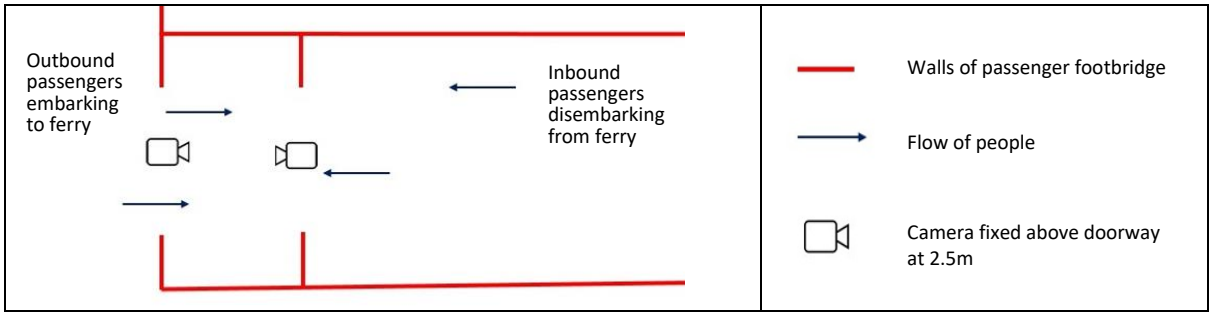


Figure 12 — Port of Hull: Layout



Terminal 1 Inbound | *Terminal 1 outbound*

Car lane 1 | *Freight passport lane 6*

Figure 1 — Port of Hull: Example footage

B.5 Stratford Westfield 28 June 2018 and 26 July 2018

Deployment details			
28 June 2018	27 / 22 °C Scattered clouds.	Environment	Free flowing – no control
Duration	6 hrs 00 min		
Crowd throughput	1670 per hour (estimate)	Watchlist size	489

Outcomes			
Crowd		**Bluelist**	
# Recognition opportunities	10000(estimated)	# Recognition opportunities	36
# Alerted	5	# Correctly Alerted	29
# Engaged	1		
# Alert confirmed correct	0		
# Action	-		

Table 10 – Stratford Jun 2018: Deployment details and LFR Outcomes

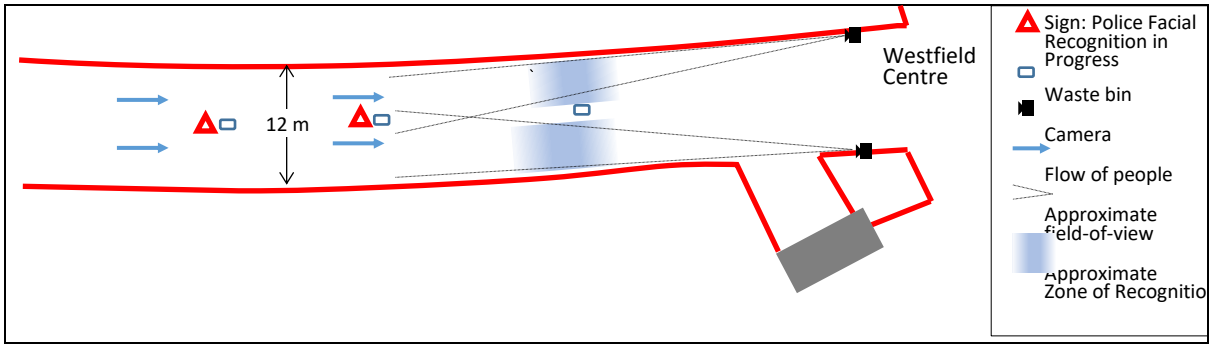


Figure 14 — Stratford Jun/July 2018: Layout



Figure 15 – Stratford July 2018: Example footage

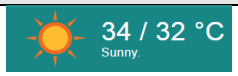
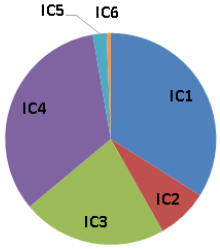
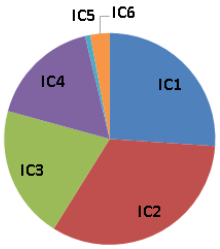
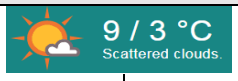
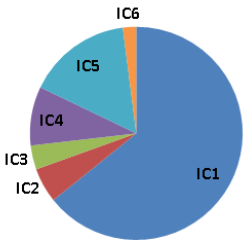
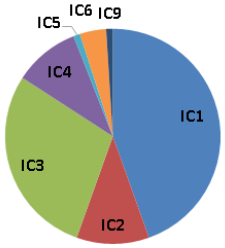
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<tr><th colspan="4">Deployment details</th></tr>
<tr><td colspan="2">26 July 2018 
Duration 5 hrs 15 min</td><td colspan="2">Environment Free flowing – no control</td></tr>
<tr><td colspan="2">Crowd throughput: 2556 per hour (estimate)
 
Crowd: perceived ethnicity</td><td colspan="2">Watchlist size: 306
 
Watchlist: perceived ethnicity</td></tr>
<tr><th colspan="4">Outcomes</th></tr>
<tr><th colspan="2">Crowd</th><th colspan="2">Bluelist</th></tr>
<tr><td># Recognition opportunities</td><td>13400 (estimate)</td><td># Recognition opportunities</td><td>62</td></tr>
<tr><td># Alerted</td><td>1</td><td># Correctly Alerted</td><td>45</td></tr>
<tr><td># Engaged</td><td>1</td><td></td><td></td></tr>
<tr><td># Alert confirmed correct</td><td>0</td><td></td><td></td></tr>
<tr><td># Action</td><td>-</td><td></td><td></td></tr>
</table>

Table 11 - Stratford July 2018: Details of deployment and LFR Outcomes

B.6 Soho 17 and 18 December 2018

Deployment details			
17 December 2018 			Environment Free flowing – no control
Location	*Cambridge Cir*	*Cranbourn St*	
Duration	3 hr 50 min	1 hr 55 min	
Crowd throughput	330 per hour	1500 per hour	Watchlist size: 2226
 Crowd: perceived ethnicity			 Watchlist: perceived ethnicity

Outcomes			
Crowd		**Bluelist**	
# Recognition opportunities	4100 (estimate)	# Recognition opportunities	43
# Alerted	5	# Correctly Alerted	32
# Engaged	3		
# Alert confirmed correct	1		
# Arrest/Action	2		

Table 12 – Soho 17 Dec 2018: Deployment details and LFR outcomes

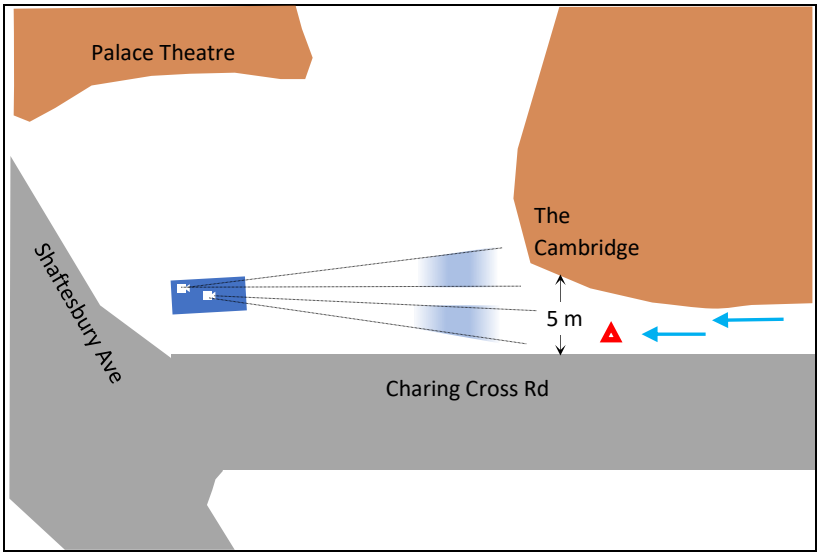
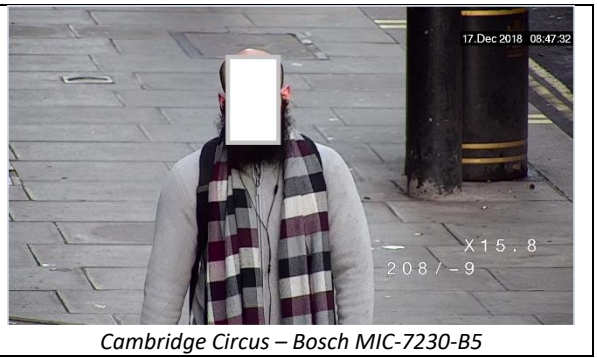


Figure 16 – Soho 17 Dec 2018: Layout Cambridge Circus



Cambridge Circus – Axis LE 8685



Cambridge Circus – Bosch MIC-7230-B5

Figure 17 – Soho 17 Dec 2018: Example footage Cambridge Circus

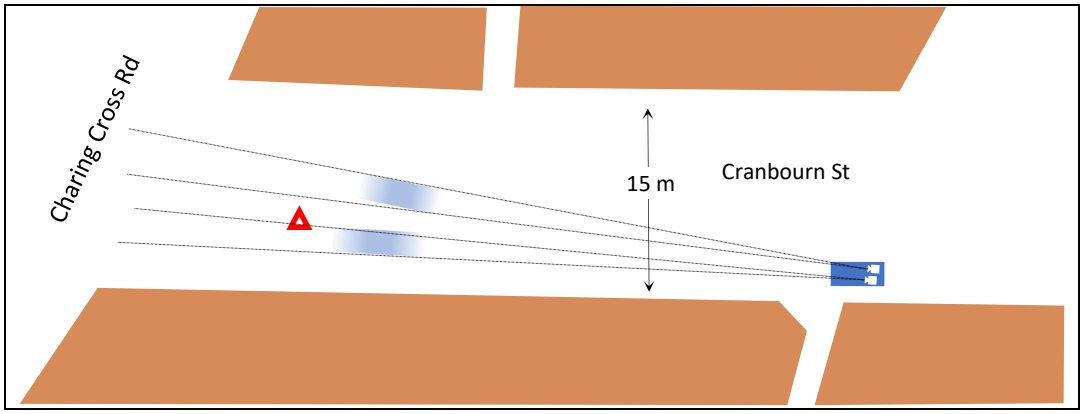
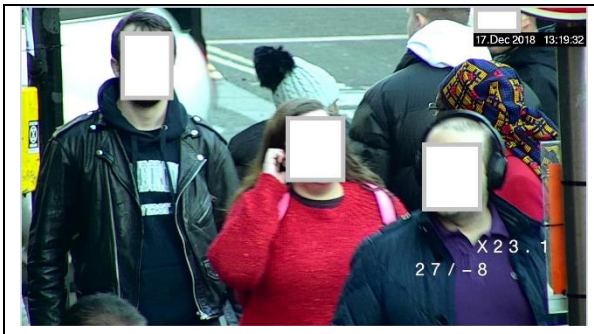


Figure 18 – Soho 17&18 Dec 2018: Layout Cranbourn St



Cranbourn St – Bosch MIC-7230-B5



Cranbourn St – Axis LE 8685

Figure 19 – Soho 17 Dec 2018: Example footage

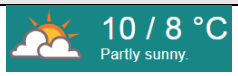
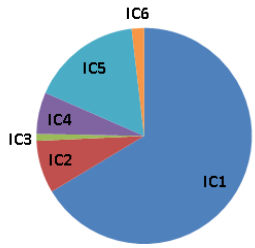
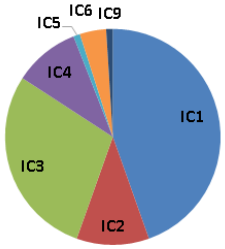
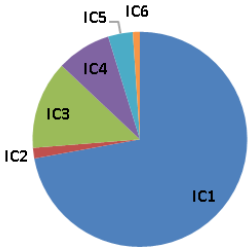
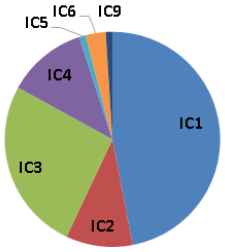
Deployment details	
18 December 2018 	Environment Free flowing – no control
Duration 5 hr 38 min	
Crowd throughput 1500 per hour (estimate)	Watchlist size: 2226
 Crowd: perceived ethnicity	 Watchlist: perceived ethnicity
Outcomes	
Crowd	**Bluelist**
# Recognition opportunities 8400 (estimate) # Alerted 9 # Engaged 1 # Alert confirmed correct 1 # Arrest/Action 1	# Recognition opportunities 58 # Correctly Alerted 45

Table 13 – Soho 18 Dec 2018: Deployment details and LFR outcomes

B.7 Romford 31 January and 14 February 2019

Deployment details	
31 January 2019  Duration 7 hr 10 min	Environment Free flowing – no control
Crowd throughput 1020 per hour (estimate)  Crowd: perceived ethnicity	Watchlist size: 2401  Watchlist: perceived ethnicity
Outcomes	

Crowd		**Bluelist**	
# Recognition opportunities	7300 (estimate)	# Recognition opportunities	70
# Alerted	10	# Correctly Alerted	46
# Engaged	5		
# Alert confirmed correct	3		
# Arrest/Action	2		

Table 14 – Romford January 2019: Deployment details and LFR outcomes

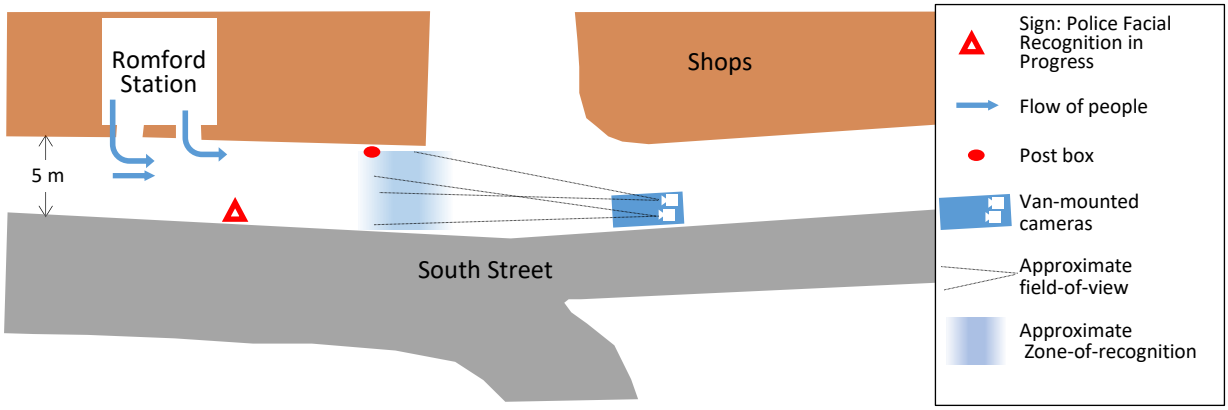


Figure20: Romford Jan/Feb 2019 – Layout



Figure21: Romford Jan: Example footage - Adjoining field of view of the two cameras

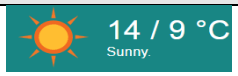
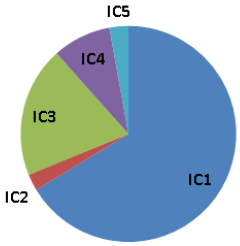
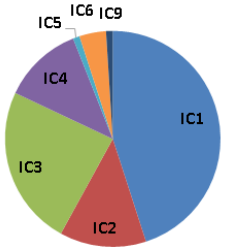
Deployment details			
14 February 2019 		Environment	Free flowing – no control
Duration	6 hr 45 min		
Crowd throughput	1500 per hour (estimate)	Watchlist size:	1996
 Crowd: perceived ethnicity		 Watchlist: perceived ethnicity	
Outcomes			
Crowd		**Bluelist**	
# Recognition opportunities	10100 (estimate)	# Recognition opportunities	70
# Alerted	16	# Correctly Alerted	48
# Engaged	8		
# Alert confirmed correct	3		
# Arrest/Action	3		

Table 15 – Romford February 2019: Deployment details and LFR outcomes



Figure 22 — Romford Feb 2019: Example footage showing adjoining field of view of the two cameras

Annex C: Face detection rate

The first step of facial recognition is detection of faces in the video. If the LFR system does not detect a face, then there will be no attempt at comparison to the watchlist, and no chance of an alert being generated, such cases contributes to the false negative identification rate.

The **Face Detection Rate** is calculated as $\frac{\text{Num. recognition opportunities with a face detected}}{\text{Num. recognition opportunities}}$

The Neoface system can be configured to show a bounding box around the faces that have been detected in the video frames as they are displayed on the system console. This can be used to estimate the face detection rate. By video recording a number of short samples of the console screens throughout the deployment, these samples can be inspected to count the total number of faces appearing in the samples, and the proportion of these that have a bounding box to estimate an average face detection rate.

The face detection rates for the first six deployments are shown in Table 16.

The face detection rate was not calculated for the Soho or Romford deployments, as the system was not always showing the bounding boxes from face detection. Moreover, due to the computational workload of real-time face detection when there are several faces in the image, there is not time for the software to display the boxing before moving on to the next video frame. It should also be noted that parameter settings in the LFR system can limit the number of faces to be detected in a video frame. This limit was probably reached in the deployments with high crowd throughput and wide field of view, but raising the limit resulted in the system skipping frames in order to maintain real-time processing. (This was also noted in "An Evaluation of South Wales Police use of automated facial recognition" [22])

Deployment	Algorithm Version	Face Detection Rate
Notting Hill 2016	S17	70%
Notting Hill 2017	S17	33%
Remembrance Sunday 2017 Parliament Sq	S17/M20	80%
Hull June 2018 Terminal 1 Inbound	M20	72%
Hull June 2018 Terminal 1 Outbound	M20	68%
Hull June 2018 Vehicle Lane 1	M20	54%
Hull June 2018 Freight Passport Lane 6	M20	25%
Hull June 2018 Rapid Deployment	M20	85%
Stratford Jun 2018	M20	71%
Stratford Jul 2018	M20	75%

Table 16 - Face detection rate by deployment

Face detection rates were typically in the range 70-80%, the exceptions being:

- Notting Hill Carnival 2017, which had high crowd throughput and wide field of view
- The vehicle lanes at Hull 2018, where the cameras for drivers in cars, vans, lorries and motorcycles, were not well located to get a reasonable quality frontal facial image.

Annex D. Poster providing information on the LFR Trial

Police Trial of Live Facial Recognition Technology

The Metropolitan Police Service (MPS) is currently trialling the use of facial recognition technology in the identification of specific individuals at various events and locations.



The trial will involve the use of cameras and facial recognition technology that are set up solely for the duration of a specific deployment.

The camera scans the faces of people as they pass by and alerts police officers to potential matches of specific individuals.

The deployment of these cameras and the watch list will be intelligence led and temporary. Only images which alert against a watch list subject will be retained for a limited period.

The use of facial recognition technology aims to support standard policing activity to ensure everyone's safety.

More information about this trial can be found at:
https://www.met.police.uk/live-facial-recognition-trial/

If you have any views or queries in relation to this trial you can e-mail us at:

BiometricFacial.Recognition@met.police.uk

METROPOLITAN POLICE

NEW SCOTLAND YARD

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